FedIRT-DP with Contaminated Response Screening

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Overview

The FedIRT-DP model developed by Zhou, Luo & Ji (2025) is a Federated Learning approach applied to Item Response Theory (IRT) models with an added student-level Differential Privacy (DP) layer to further obfuscate adversarial attempts to reverse-engineer student data. On top of effectively anonymizing student data, the DP layer is also shown to be capable of mitigating the impact of extreme response rows, as shown in Study 3.

The primary motivation behind this report was how FedIRT, despite its robustness against extreme response rows with DP, possesses no counter-measures against such rows. This motivated me to look into the potential benefits of an implementation that could identify such response rows from the individual student gradients.

A brief initial check proved that Local Outlier Factor (LOF) and Angle-Based Outlier Detection (ABOD) algorithms were able to identify the rows with extreme responses. However, before further examination, two preceding questions had to be addressed: Is the impact of extreme response rows significant enough to risk false positives from an additional statistical model to detect them? And how does FedIRT fare against other forms of contaminated responses?

Stress-Testing FedIRT Against Inaccurate Responses

Procedure

Staying faithful to the study, all parameters were followed if explicitly stated; by the very least inferred if the paper implied so. Specifically, school effect was set to $s_k = 0$, with each of the 10 schools ($K = 10$) having 100 students each ($N_k = 100$). The test given to each school was dichotomously scored with 10 items ($J = 10$),

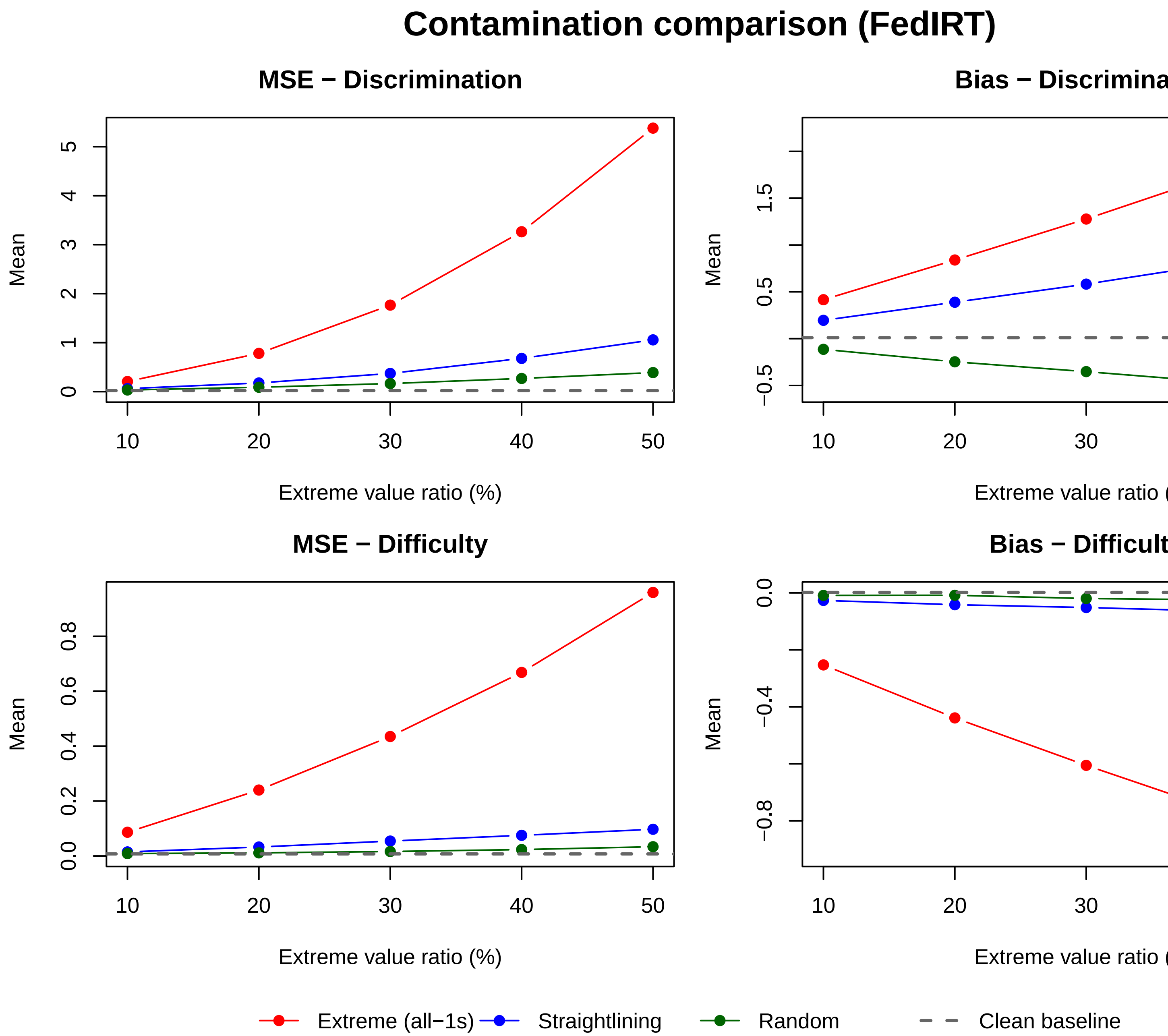
Individual abilities for each student were generated from a standard normal distribution ($\theta_i \sim N(0, 1)$) and fixed. Discrimination parameters were drawn from a uniform distribution ranging from $\alpha \in [0.5, 2]$, and difficulty parameters were drawn from a uniform distribution ranging from $\beta \in [-1, 1]$ to encompass both low and high parameter conditions. Student responses were generated using the 2PL Model probability using these fixed parameters.

The contamination conditions examined were straightlining (a certain number of items in a row are answered the same), and random reporting. Extreme response rows (all-ones) was also included to ensure the setup was consistent with the original study. As inferred from the plots in Study 3, a range of contaminated response proportions were tested from $[0.1, 0.2, ..., 0.5]$ for each contamination condition. Each proportion and each condition was repeated 100 times ($T = 100$).

The same MSE and bias formulas (Eq. 26) were used from the study for each proportion and contamination condition.

Results

The behavior of the MSE and bias of the difficulty and discrimination parameters are consistent with those from the original study for the extreme response condition. MSE and bias tend to increase as the proportion of responses consisting of all ones increase, except for the difficulty parameter which slightly decreases.



The MSE and bias of the other conditions, straightlining and random reporting, tend to follow the same positive relationship with the proportion of inaccurate responses but to a lesser degree. An exception to this is the bias of the discrimination parameter for random reporting, which steadily decreases below the baseline levels.

Table 1: MSE by contamination type and proportion

	Discrimination (a)				**Difficulty (b)**			
Prop (%)	Clean	Extreme	Straight	Random	Clean	Extreme	Straight	Random
10	0.021	0.207	0.061	0.034	0.007	0.086	0.015	0.009
20	0.021	0.781	0.177	0.089	0.007	0.240	0.033	0.011
30	0.021	1.766	0.372	0.165	0.007	0.435	0.054	0.016

40	0.021	3.264	0.678	0.269	0.007	0.668	0.075	0.023
50	0.021	5.380	1.057	0.389	0.007	0.960	0.097	0.034

Table 2: Bias by contamination type and proportion

	Discrimination (a)				**Difficulty (b)**			
Prop (%)	Clean	Extreme	Straight	Random	Clean	Extreme	Straight	Random
10	0.011	0.416	0.196	-0.114	0.001	-0.253	-0.027	-0.009
20	0.011	0.840	0.389	-0.247	0.001	-0.439	-0.042	-0.008
30	0.011	1.277	0.582	-0.352	0.001	-0.605	-0.051	-0.020
40	0.011	1.749	0.800	-0.464	0.001	-0.762	-0.064	-0.024
50	0.011	2.248	1.003	-0.566	0.001	-0.923	-0.076	-0.028

Anomaly Detection to Identify Inaccurate Responses from Per-Student Gradients

Procedure

The same conditions and parameters were used, except for the number of schools and the proportion of contaminated rows. LOF and ABOD algorithims were fit on the 100 by 10 response matrices of a single school across different contamination conditions. The proportion of contaminated rows was fixed at 0.3.

Performance was measured using AUC given the objective was to identify inaccurate responses, and identified rows could be compared against their true labels.

Results

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# table of AUC
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Discussion and Future Directions